



Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510

Fax: +66-2721-3515

SAFETY DATA SHEET

Company Identification

Thep Watana Co., Ltd. Address: 382 Bangpoo Industrial Estate.

Sukhumvit Road, Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66(0)-2721-3510;

Fax: +66(0)-2721-3515

1. CHEMICAL IDENTIFICATION

Product Identification

Product name: propineb 70% WP

Product use: Fungicide

Relevant identified uses of the substance/mixture and uses advised against:

Used as a protective treatment on several crops for the control of various fungi, especially Oomycetes, Ascomycetes, Basidiomycetes and Fungi imperfect. Propineb controls blight on potatoes and tomatoes, downy mildew on hops and vines, apple scab, blue mould on tobacco and Sigatoka disease of bananas. It can also be used on gooseberries, black currants, celery and cereals.

2. HAZARD IDENTIFICATION

More important danger for the man: None

Health Effects: No adverse health effects are expected if this product is used in accordance with the label. Alcohol greatly increases the toxicity of this product. Do not consume alcohol on the day of use. Repeated exposure may cause allergic disorders. May irritate the eyes.

3. COMPOSITION/INFORMATION ON INGREDIENT

Chemical name: 1-(dithiocarboxyamino)propan-2-ylcarbamodithioic acid

Common name: propineb

Formula: $C_5H_{10}N_2S_4$

Molecular weight: 226.4 g/mol



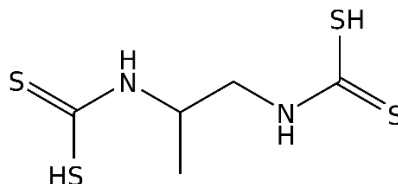
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Structural formula:



Ingredients and composition:

Component	CAS No.	%W/W
Active ingredient		
- propineb	12071-83-9	70
Other ingredients		30

4. FIRST AID MEASURES

Description of first aid measures

In case of swallowed:

Call a poison control center or doctor immediately for treatment advice. Have person sip a glass of water if able to swallow. Do not induce vomiting unless told to do so by a poison control center or doctor. Do not give anything by mouth to an unconscious person.

In case of eyes contact:

Hold eye open and rinse slowly and gently with water for 15 – 20 minutes. Remove contact lenses, if present, after the first 5 minutes, then continue rinsing eye. Call a poison control center or doctor for treatment advice.

In case of skin contact:

Take off contaminated clothing. Rinse skin immediately with plenty of water for 15 – 20 minutes. Call a poison control center or doctor for treatment advice.

In case of Inhaled contact:

Move person to fresh air. If person is not breathing, call an ambulance, and then give artificial respiration. Preferably mouth-to-mouth if possible. Call a poison control center or doctor for further treatment advice.

Note to Physician

Apply basic aid and decontamination procedures. Treat symptomatically.

Antidote: none known



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5. FIRE FIGHTING MEASURES

Extinguishing Media:

Water fog or foam

Fire and explosion hazards:

If involved in a fire, oxides of nitrogen and sulfur should be expected.

Measure of personal protection:

Firefighters should wear NIOSH/MSHA approved self-contained breathing apparatus and full protective clothing. Dike areas to prevent runoff and contamination of water sources. Dispose of fire control water later.

6. ACCIDENTAL RELEASE MEASURES (SPILLAGE)

Personal cautions:

Prevent spill from spreading or entering waterways or drains. Vacuum or sweep up spilt material. Fill material along with any contaminated soil etc. into containers.

Cleaning methods

Clean affected area with an aqueous detergent and a small amount of water. Absorb this with hydrated lime and place in a sealable drum. On completion of clean-up remove and wash all protective clothing and equipment with detergent and water. Any heavily contaminated clothing should be placed in a plastic garbage bag and placed in a sealable drum. Do not smoke, eat or drink during the clean-up operation.

7. HANDLING AND STORAGE

Good industrial practice in housekeeping and personal hygiene should be followed.

Safe Handling:

Do not breathe dust. Avoid direct or prolonged contact with skin and eyes. Wash hands and face carefully before eating, drinking, using tobacco, applying cosmetics, or using the toilet. Wash exposed skin promptly to remove accidental splashes of contact with this material.

Storage Conditions:

Keep out of reach of children. Store in the tightly closed original container in a dry, well-ventilated area, as cool as possible. Do not store for prolonged periods in direct sunlight. Do not store near any material intended for use or consumption by humans or animals.



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8. EXPOSURE CONTROL/PERSONAL PROTECTION

Personal Protective Measures

Avoid contact with eyes, skin and clothing during mixing/pouring operations. Ensure adequate ventilation. Do not inhale spray mist or vapor.

Personal Protective Equipment

When handling concentrate and preparing spray wear overalls, boots, elbow-length PVC gloves, and half face respirator with dust cartridge.

Industrial hygiene

If clothing becomes contaminated with product or wet with spray, remove clothing immediately. If product or spray on skin, immediately wash the area with soap and water. After use and before eating, drinking or smoking, wash hands, arms and face thoroughly with soap and water. After each day's use wash contaminated clothing, gloves etc. with detergent and water.

9. PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Physical stage:	Powder
Color:	White
Odor:	Weak characteristic odor
Flammability:	Not flammable
Vapor pressure:	Not available
Density:	Not available
Solubility in water:	Dispersible
Stability:	Stable in normal conditions

10. STABILITY AND REACTIVITY

Conditions to avoid:	Not applicable
Chemical stability:	This material is stable under normal handling and storage conditions.
Incompatibility:	Strong bases, strong acids, strong oxidizing agents
Products to avoid:	Not applicable
Thermal decomposition:	Not applicable
Hazardous decomposition products:	Oxides of nitrogen and sulfur
Hazardous reaction:	Will not occur



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11. TOXICOLOGICAL INFORMATION

Information on Acute toxicity for end product, Propineb:

LD50 (Oral, rat):	> 5,000 mg/kg
LD50 (Dermal, rat):	> 5,000 mg/kg
LC50 (Inhalation, rat, 4h):	> 2.0 ml/L
Skin irritation (rabbit):	Slight irritation to skin of rabbit
Eye irritation (rabbit):	Slight irritation to eyes
Skin sensitization:	No-sensitive to guinea pigs

12. ECOLOGICAL INFORMATION

Toxicity

Adopt good working practices, so that the product is not released into the environment.

Data referred to the mixture, propineb:

Fish – Acute toxicity:

Rainbow trout,

LC50 (96 h):	1.9 mg/L
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Golden orfe,

LC50 (96 h):	133 mg/L
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Invertebrates – Acute toxicity:

Water flea (Daphnia magna),

LC50 (48 h):	4.7 mg/L
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Birds – Toxicity:

Japanese quail,

LC50 (5 days):	> 5,000 mg/kg Dietary
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Algae – Toxicity:

Algae,

ErC50 (96 h):	2.7 mg/L
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Arthropod – Toxicity:

Bee,

LD50 (oral):	> 100 ug/bee	Not toxic to bees
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13. DISPOSAL CONSIDERATIONS

After intended use:

Completely empty bag into pre-mix bucket. Single rinse before disposal. Add rinsing to the spray tank. Do not dispose of undiluted chemicals on site. Puncture and bury empty container in a



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local landfill (check statutory requirements). If not available, bury the container below 500 mm in a marked disposal pit specifically set up for this purpose, clear of waterways, vegetation, and roots. Empty containers and products should not be burnt.

After spill or accident:

Dispose of sealed containers at an approved local waste disposal site.

14. TRANSPORT INFORMATION

The data provided in this section is for information only. Please apply the appropriate regulations to properly classify your shipment for transportation.

UN No: 3077

Dangerous Goods Class: 9

Packaging Group: III

15. REGULATORY INFORMATION

US Federal

TSCA 12b export notification:

None

SARA Title III – section 302 – notification and information:

None

SARA Title III – section 313 – toxic chemical release reporting:

None

Environmental

CERCLA:

None

Clean Water Section 307 Priority Pollutants:

None

Safe Drinking Water Act Maximum Contaminant Levels:

None

16. OTHER INFORMATION

All information and instructions provided in this SDS are based on the current state of scientific and technical knowledge at the date indicated on the present MSDS and are presented in good faith and believed to be correct. This information applies to the product as such. In case of formulations or mixes, it is necessary to ascertain that a new danger will not appear. It is the responsibility of persons in receipt of this SDS to



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ensure that the information contained herein is properly read and understood by all people who may use, handle, dispose or in any come in contact with the product. If the recipient subsequently produce formulations containing this product, it is the recipients sole responsibility to ensure the transfer of all relevant information from this SDS to their own SDS.

THEP WATANA CO., LTD.